Expert Interview 1 — Karim Kazhymukan Baimuratuly

Position: IoT Specialist and STEM Teacher, Judge and Organizer of International STEM Olympiads

Date: 21 October 2026

Q1. Is ESP32 a suitable choice for a water-level monitoring prototype?

Answer: Yes, I think ESP32 is a good choice for an educational IoT project. It has built-in Wi-Fi, is relatively affordable, and can communicate with a backend server without requiring many additional components. However, before assembling the system, the team should confirm the exact board model, its power requirements, and compatibility with the selected sensor.

Q2. What should the team consider when selecting and connecting a water-level sensor?

Answer: The sensor must match the measurement range and the conditions in which it will operate. If the team chooses an ultrasonic sensor, it should consider the minimum measurement distance, mounting position, and possible interference from the water surface or surrounding objects. Electrical compatibility is also important. ESP32 uses 3.3 V logic, so the sensor's output voltage must be checked before connecting it to the board.

Q3. How can the team make sure that the sensor readings are accurate?

Answer: I would recommend testing the sensor at several known water levels and comparing its readings with reference measurements. Each measurement should be repeated so the team can identify errors and inconsistent results. Filtering can help reduce random fluctuations, but it should not hide a genuine rapid increase in water level. The team should keep the original readings as well, so the results can be checked later.

Q4. What should happen if the Wi-Fi connection is interrupted?

Answer: The device should detect unsuccessful requests and handle reconnection automatically. If the connection returns, the system should be able to identify which measurements are new and which have already been received. Sequence numbers or

unique identifiers can help prevent duplicate records. I would also recommend testing both the disconnection and recovery scenarios rather than testing only normal operation.

Q5. Which tests would you consider essential before demonstrating the prototype?

Answer: I would test normal measurements, a rising water level, threshold crossings, invalid sensor readings, Wi-Fi disconnection, backend unavailability, and reconnection. The team should record the expected and actual results for each scenario. It is also important to check what the system displays when measurements are missing, because the interface must not confuse a communication problem with a safe water level.

Q6. Is it necessary to implement water-level forecasting in the first version?

Answer: I would treat forecasting as an additional feature rather than a core requirement. First, the team needs a reliable measurement and data-processing pipeline. After that, it can test a simple calculation based on recent water-level changes. The estimate should be shown only when there are enough valid measurements and the trend is reasonably stable. A machine-learning model would require suitable data and additional evaluation, so I would not make it a priority at this stage.

Expert Interview 2 — Tungatarov Erzhigit Erbatyrovich

Position: Executive Director of Kokshetau Zhylu

Date: 21 October 2026

Q1. From an operational perspective, what makes a monitoring system useful?

Answer: The information must be clear, timely, and useful for making decisions. A person responsible for monitoring should be able to see the current measurement, understand whether it requires attention, and know when the data were last updated. It is also important to identify which device provided the information. A system is less useful if employees have to spend extra time figuring out whether its readings can be trusted.

Q2. What limitations should the team consider when developing a small monitoring prototype?

Answer: A prototype tested in a controlled environment is different from a system installed at an actual utility site. In practice, the team would need to consider environmental conditions, power supply, communication reliability, maintenance, sensor suitability, and the procedures followed when an alarm occurs. These factors cannot be fully evaluated through a laboratory demonstration alone. The prototype should therefore have a clearly defined scope and should not be presented as a ready-to-use industrial solution.

Q3. How should the system respond when measurements become unavailable or unreliable?

Answer: It should clearly indicate the problem instead of continuing to display an old value as if it were current. The last valid measurement can remain available for reference, but its timestamp and outdated status must be visible. The system should also distinguish between a device that has lost communication and a device that is still connected but producing invalid measurements. These situations require different responses.

Q4. Would you recommend including automatic physical pump control in the first version?

Answer: I would not make automatic pump control part of the initial demonstration. Before controlling physical equipment, the team would need to verify the hardware, define safety mechanisms, and test how the system behaves during communication failures and incorrect measurements. For an educational prototype, a simulated pump state is sufficient to demonstrate the intended workflow without introducing unnecessary operational risks.

Q5. What would be required before considering the system for use at a real facility?

Answer: The system would need repeatable test results demonstrating measurement accuracy, correct warning behaviour, and reliable handling of communication failures and recovery. The sensor and installation method would also need to be assessed for the intended environment. In addition, there should be a clear procedure for responding to warnings and maintaining the equipment. Only after these checks could the team make a more informed assessment of its suitability for practical use.

Expert Interview 3 — Abdullayev Shynggys

Position: Front-End Developer with Three Years of Experience

Date: 21 October 2026

Q1. What should an operator understand first when opening the dashboard?

Answer: The operator should immediately understand the current water-level situation without having to search through different screens. I would place the latest valid measurement, its unit, the current status, and the time of the last valid reading together in the main dashboard area. The warning state should be immediately noticeable. A graph can provide additional context, but it should not distract from the most important information.

Q2. What information should be available beyond the current water-level reading?

Answer: I would include a historical chart and an event log. The chart helps the operator understand how the water level changes over time, while the event log shows when warnings appeared and when the device lost communication. It is important to display timestamps and units consistently. If measurements are missing, the chart should show a visible gap instead of connecting the available points and suggesting that data were recorded continuously.

Q3. How should the interface display warning and critical states?

Answer: I would use clear text labels such as WARNING and CRITICAL, supported by distinct visual indicators. Colour can help users recognise the states quickly, but it should not be the only way to communicate them. The interface should remain understandable on smaller screens and for users who have difficulty distinguishing colours. The most important warning information should be visible without requiring the operator to open another page.

Q4. What should the dashboard show when the device stops sending data?

Answer: It should clearly display an OFFLINE or STALE DATA status, depending on the situation. The system can preserve the last valid measurement, but it must show when that measurement was recorded. For example, if the last reading was low but the device has stopped communicating, the dashboard should not continue presenting

that value as proof that the current water level is safe. The interface must distinguish the last known condition from the current device status.

Q5. Would a forecast showing the estimated time until a critical water level be useful?

Answer: It could be useful because it gives the operator additional context about how quickly the water level is rising. However, I would display it as an approximate estimate rather than a guaranteed prediction. If there are too few measurements, the data are outdated, or the trend is unstable, the interface should explain why the estimate is unavailable. The actual measured level and warning state should always remain the primary information.